

Math 432/532: Introduction to Topology II

Solutions to HW #8

1. Suppose that $\{K_r \mid r \geq 0\}$ is a filtered complex. More precisely, each K_r is an abstract simplicial complex with the same vertex set, and $K_{r_1} \subset K_{r_2}$ whenever $r_1 \leq r_2$. Define r to be a **critical scale** if, for all $\epsilon > 0$, $K_{r-\epsilon} \subsetneq K_r$.

- (i) Let $X = \mathbb{R}^2$ with its Euclidean metric, and let $S = \{(1, 0), (0, 1), (-1, 0), (0, -1)\} \subset X$. Determine all of the critical scales for $\{\text{Rips}(S, r) \mid r \geq 0\}$ and describe the complex at all parameters.

Solution: The critical scales are $\sqrt{2}$ and 2. When $r < \sqrt{2}$, $\text{Rips}(S, r)$ is discrete. When $\sqrt{2} \leq r < 2$, it is a square. When $r \geq 2$, it is a tetrahedron.

- (ii) With the same X and S as in part (i), determine all of the critical scales for $\{\text{Cech}(S, r) \mid r \geq 0\}$ and describe the complex at all parameters.

Solution: The critical scales are $\frac{\sqrt{2}}{2}$ and 1, and the three different complexes that you can get are the same as those in part (i).

- (iii) Redo part (ii) with the same set S but this time with $X = S^1$, still with the Euclidean metric.

Solution: This is the most interesting one! Now the critical scales are $\sqrt{2 - \sqrt{2}}$, $\sqrt{2}$, and $\sqrt{2 + \sqrt{2}}$. When $r < \sqrt{2 - \sqrt{2}}$, $\text{Cech}(S, r)$ is discrete. When $\sqrt{2 - \sqrt{2}} \leq r < \sqrt{2}$, it is a square. When $\sqrt{2} \leq r < \sqrt{2 + \sqrt{2}}$, it consists of all proper faces of a tetrahedron. When $r \geq \sqrt{2 + \sqrt{2}}$, it is the full tetrahedron.

Note that the somewhat mysterious number $\sqrt{2 - \sqrt{2}}$ is the Euclidean distance from $(1, 0)$ to $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, while $\sqrt{2 + \sqrt{2}}$ is the distance from $(-1, 0)$ to $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.

2. Let X and S be as in problem 1(i), and let

$$T = \left\{ \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right), \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right) \right\} \subset X.$$

Compute the minimum $\epsilon \geq 0$ such that $\{\text{Rips}(S, r) \mid r \geq 0\}$ and $\{\text{Rips}(T, r) \mid r \geq 0\}$ are ϵ -interleaved.

Solution: This is sort of a trick question. In fact, $\text{Rips}(S, r)$ and $\text{Rips}(T, r)$ are isomorphic via the morphism that rotates the plane 45 degrees clockwise, which means that the two filtrations are 0-interleaved.